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A PALEOMAGNETIC STUDY OF CERTAIN MESOZOIC FORMATIONS IN NORTHERN MEXICO

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Rocks of Late Cretaceous, Early Jurassic and Late Triassic age, collected in northern Mexico yield the following pole positions: 169.3°E57.9°N (Cretaceous), 70.7°E76.0°N (?Jurassic) and 119.2°E76.4°N (?Late Trassic). The Triassic and Cretaceous poles are not significantly different from those class-A poles (Hicken et al., 1972) of the North American craton. It is therefore suggested that the North American craton may be traced south as far as 23°N and inferentially a further four degrees (to the Mexican volcanic belt).

The results from the La Boca Formation are interpreted as indicating a much greater age (Late Precambrian–Early Paleozoic) than is currently assigned to that formation.

1. Introduction

The origin of the Gulf of Mexico and of the Caribbean are problems of major geophysical interest. The significance of the bordering lands of Central America, clearly demonstrated by the overlap of Central America onto South America in the reconstructions of the Atlantic by Bullard et al. (1965), has resulted in a num-

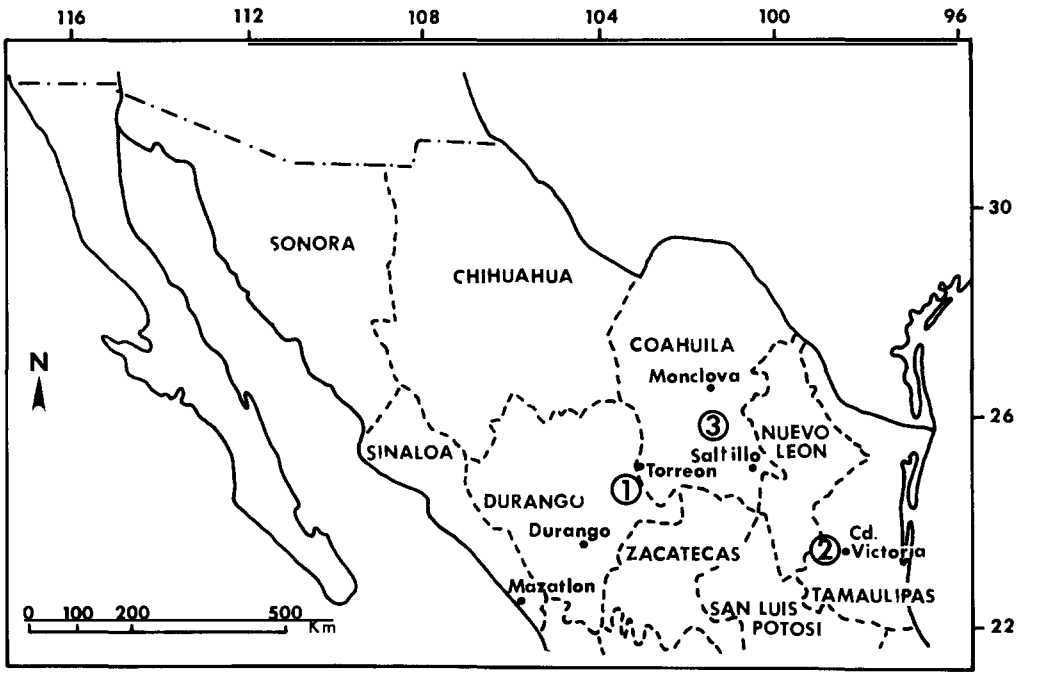


Fig. 1. Outline map of Mexico illustrating the location of the principal sections sampled. *1* = Upper Triassic, *2* = Lower Jurassic, *3* = Upper Cretaceous.

ber of ingenious hypotheses such as that of Freeland and Dietz (1971). Paleomagnetism can contribute to the question of the origin of the Caribbean, for plate motion close to its pole of rotation will result in large displacements of the virtual pole positions of rocks located on that plate. There has already been some use of paleomagnetism in the general area of the Caribbean (Watkins and Cambray, 1971; Hargraves and Shagam, 1969; MacDonald and Opdyke, 1972) and the results indicate large-scale displacements. The present paper presents the results of the first stage of an investigation of the western margin of the Gulf of Mexico and of the Caribbean, the investigation of Mesozoic rocks north of the 19th parallel. The results of the study of Cenozoic rocks of the same region have been presented elsewhere (Nairn et al., 1975).

Rocks have been sampled in the states of Durango, Coahuila and Tamaulipas (Fig. 1). Most of the outcrops are in the eastern part of the country within the Sierra Madre Oriental for the area to the west is heavily blanketed by Neogene ash-flow tuffs and associated volcanic products. The underlying Paleozoic and Precambrian rocks, where exposed, belong to the Oaxacan or Huastecan structural belts (De Cserna, 1970, 1971).

2. General geology

All the beds examined have been involved in the fold movements which led to the formation of the Sierra Madre Oriental. In the east, in Tamaulipas the rocks lie in broad open folds, and numerous outcrops of red beds in the region of the Miquihuana arch have been reported (Baker, 1971). Tighter folding with low-angle overthrusting is reported in Coahuila along the northern margin of the Parras basin, but the rocks are, on the whole, relatively undeformed and essentially unmetamorphosed. Red beds have been attributed to each of the Mesozoic Systems, but characteristically unfossiliferous, correlation of these beds is at best tenuous and often invalid. A general description of the sedimentation and paleogeography is given by De Cserna (1970).

In many parts of northern and eastern Mexico Upper Jurassic (Oxfordian) limestones unconformably overlie a continental red-bed sequence. In the east, in Tamaulipas near Ciudad Victoria and in the region of the Miquihuana arch (Baker, 1971), the red beds were assigned to the Jurassic Huizachal Formation (Imlay et al., 1948). It has been suggested that these red beds are an onshore facies and grade eastwards into the evaporite deposits found offshore in the vicinity of the Mexican ridges in the Gulf of Mexico (Bryant et al., 1969; Wilhelm and Ewing, 1972). The beds have been correlated with the Nazas Formation in Durango (Pantoja-Alor, 1972) to which a Middle Triassic–Early Jurassic age is assigned (Clemmons and McLeroy, 1966). The Nazas Formation, however, contains volcanic rocks absent in the Huizachal Formation, and largely on the basis of the combination of red beds and volcanics, correlation with a sequence near Tezuitlan, Puebla has also been proposed.

A provisional Pb_{α} age of 230 ± 50 m.y. on the volcanic rocks in the Nazas Formation makes a Triassic age at least reasonable, but as rocks of the Huizachal Formation demonstrably overlie rocks of Early Jurassic age, correlation of Nazas and Huizachal Formations is invalid. Part of the confusion was resolved by the demonstration by Mixon et al. (1959) that the Huizachal Formation consists of two red-bed units separated by an angular unconformity. Mixon et al. (1959) introduced the names La Joya Formation for the younger and La Boca Formation for the older unit, although Fries et al. (1962) suggested retaining the name Huizachal for the older beds.

There is less confusion concerning the red beds which form part of the Difunta Group in the Parras basin near Saltillo, Coahuila. The Difunta Group is Late Cretaceous (Gulfian), i.e., Maestrichtian in age. It contains red beds at three levels of which the lower two, the Cerro Huerta Formation and the Las Imagenes Formation (Weidie et al., 1959), were sampled. The Cretaceous rocks of the Difunta Group are folded, but in the northern and northeastern parts of the Parras basin where the formations were sampled, the folds are gentle with low dips (Weidie et al., 1959).

3. Paleomagnetic study of the individual formations

Samples from all formations were cored in the field using a gasoline-powered drill. In the laboratory remanence measurements were made with a Foster fluxgate spinner in Cleveland and subsequently with a Digico spinner in Columbia. All samples were magneti-

cally cleaned; the igneous rocks of the Nazas Formation in alternating fields, the sediments by thermal demagnetization in a mu-metal room where the residual field was reduced to less than 20γ by use of an ancillary coil system.

3.1. Nazas Formation

Kellum (1936) described a Red Rock Series composed largely of lavas and interbedded shales and siltstones cropping out in Los Cerritos Colorados west-southwest of Gomez Palacios, Durango. Subsequently the unit was mapped by Clemmons and McLeroy (1966) who chose as the type the region where the Durango Highway crosses Los Cerritos Colorados even though the section is incomplete. No base is exposed, and the upper contact is marked by an erosional unconformity.

The lowest horizon described by Clemmons and McLeroy (1966) is a porphyritic-rhyolitic ignimbrite (22 m thick). This is overlain by 548 m of red-purple, medium- to thick-bedded sandstones and siltstones with mudstone partings. Lavas, so reddened that they are scarcely distinguished from the sedimentary rocks are mentioned by Kellum (1936) and some horizons are shown on the map of Pantoja-Alor (1972) which accompanies a description of the outcrop in the Mazatlan field guide. The thick unit (see Fig. 2) and two of the thinner bands were sampled.

A closely spaced sequence of samples (spacing about 50-cm intervals) of the red siltstones and sandstones was cored along the road cut through Los Cerritos Colorados, although only a third of the quoted thickness is exposed. The western end of the section is disturbed by faulting and was not sampled.

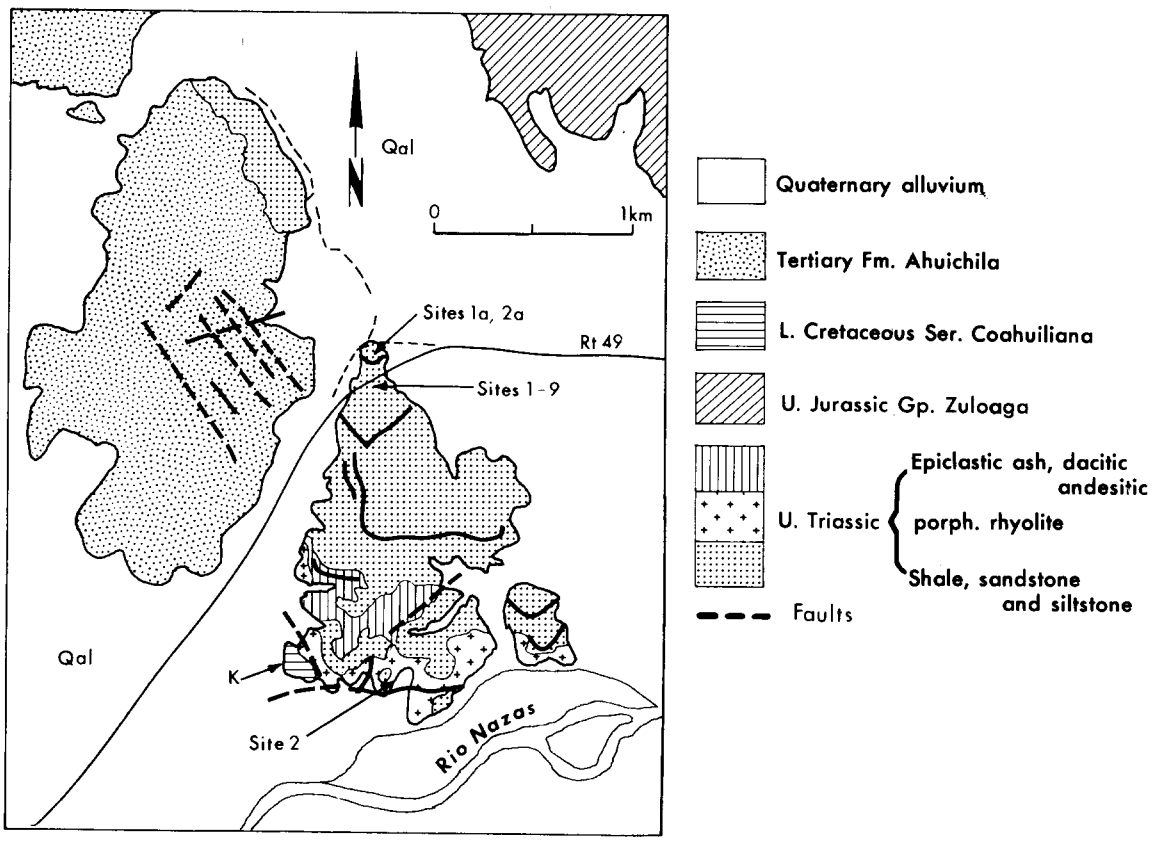


Fig. 2. Geological sketch map of Los Cerritos Colorados, near Gomez Palacios Durango, showing the outcrop of the Upper Triassic–Lower Jurassic rocks. Ignimbritic ash horizons marked by heavy lines. The location of the sites is shown. Sites *1a*, *2a* and *L* (lava) are in igneous rocks sampled in 1968, the remainder in 1972.

In general it was found necessary to heat samples to 450–500°C to remove secondary magnetic components. In all some 30–40 samples were examined. Beds were considered the recognizable stratigraphic units and individual results were combined to give a mean for a given bed. The mean was computed except where there were only two samples per bed when the mean was estimated from plots on a stereographic net. Some fourteen horizons were recognized (see Fig. 3) with a zone of reversed magnetism near the base of the exposed section. Because of vandalism in the C.W.R.U. Laboratory prior to departure, all specimens and records which would permit the construction of demagnetization figures, were lost.

3.2. Huizachal Formation

As discussed earlier the term Huizachal as a formational name is ambiguous. While in the vicinity of Cuidad Victoria, the nomenclature proposed by Mixon et al. (1959) is more precise, in other regions it is convenient to retain the older terminology "Huizachal Formation s.l." in the absence of more exact age or lithologic assignment.

Some 124 km north of Saltillo immediately south of Cuesta la Murella there is a section of nearly continuous outcrop of red beds assigned to the Huizachal Formation (E.F. McBride, personal communication, 1968). It consists of compact fine-grained sandstones and siltstones similar to those of the Nazas Formation and to certain of the lithologies of the La Boca Formation.

A suite of 29 samples, collected in a continuous sequence were thermally cleaned by heating to temperatures of 450–500°C. The directions of magnetization of the cleaned samples are shown in Fig. 4 and the data listed in Tables I and II.

3.3. La Boca Formation

It was the existence in the Caballeros Canyon of a steeply dipping series of red beds unconformably overlain by red beds with a shallow dip which led Mixon et al. (1959) to subdivide the old Huizachal Formation and to suggest a possible solution to the stratigraphic problem. A sequence of samples was collected in Los Caballeros Canyon from these steeply dipping beds

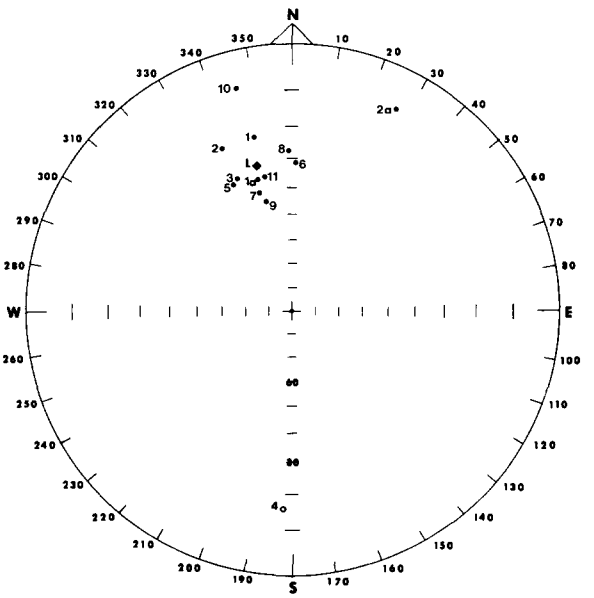


Fig. 3. Stereogram illustrating the site mean directions of magnetization after thermal demagnetization of the sediments and treatment of the volcanic rocks in alternating fields, of rocks from the Nazas Formation in Los Cerritos Colorados, Durango. The igneous rocks are identified by the sites numbered *1a, 2a,* and *L* (lava).

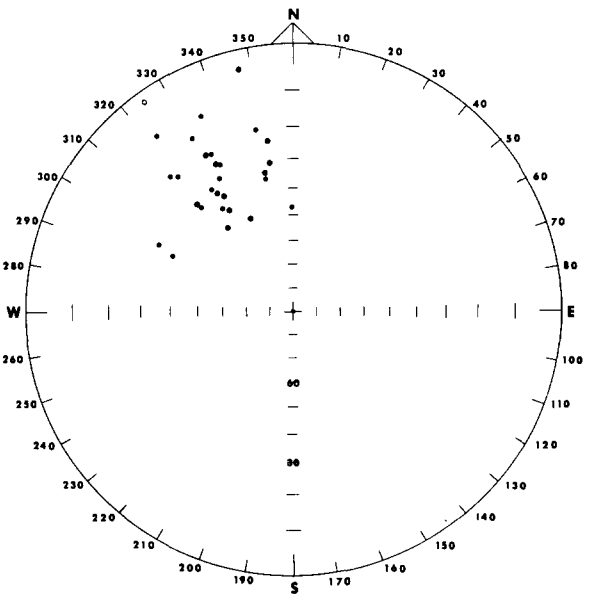


Fig. 4. Stereogram illustrating the directions of magnetization of individual cores after thermal demagnetization of rocks in the Cuesta La Murella section north of Saltillo. The rocks belong to the "Huizachal" Formation.

TABLE I

Summary of Mesozoic red-bed data

	D	I	Sampling	α_{95}	K	R
Cretaceous						
(1) Cerro Huerto section	326.0	+35.7	6 sites	10.9	3	5.869
(2) La Escondida section	215.9	−11.4	5 sites	43.2	4	4.021
(3) Sierra Guijardo section	324.5	+40.8	13 sites	13.5	10	11.843
Mean results (1) + (3)	331.7	+42.8		25.5	97	1.989
((1)−(3) inc.)	349.1	+35.1		62.5	5	2.596
Jurassic						
La Joya Formation: El Huizachal	182.5	−19.9	14 samples	19.0	5	11.562
?Jurassic						
"Huizachal" Formation: La Murella	330.4	+28.9	29 samples	5.9	22	27.722
Triassic						
Nazas Formation: Los Cerritos Colorados	350.6	+28.4	14 sites	8.2	24	13.467
?age						
La Boca Formation	222.9	−30.0	15 samples	9.6	17	14.159
Caballeros Canyon	69.3	+12.9	15 samples	15.3	7	13.039

which consisted of alternations of reddish siltstones with fine-grained grey sandstone horizons.

The rocks appeared to possess some stability as indicated by the NRM measurements (Fig. 5). The results were improved by thermal demagnetization over the range 450–550°C. The least scatter was found after cleaning at a temperature of 550°C, and the directional data are shown in Fig. 6. In view of the controversy over the time of acquisition of magnetization in red beds, it is worth recording that there was no difference in behaviour between the red and grey horizons sampled, both normal and reversed directions were found in both lithologies, and in both were samples which did not respond to thermal cleaning (see Fig. 5).

3.4. La Joya Formation

The beds sampled west of Cuidad Victoria lie immediately below the unconformity with the overlying limestone. They represent the highest horizons of the

TABLE II

Paleomagnetic pole positions from certain Mesozoic formations in northern Mexico

Age	Location	Longitude	Latitude	dm	dp
Cretaceous	(1) Sierra Guijardo	172.9E	57.6N	10.0	16.4
	(2) La Escondida	12.7E	50.4N	22.3	43.9
	(3) Cerro Huerto	165.7E	58.2N	–	–
	(1) + (2)	169.3E	57.9N	A_{95} = 8.4	
Jurassic	La Joya Formation: El Huizachal	70.7E	76.0N	10.4	19.9
?Jurassic	Huizachal Formation: La Murella	154.8E	60.6N	3.6	6.4
Triassic	Nazas Formation: Los Cerritos Colorados	119.2E	76.4N	4.9	9.0
?age	La Boca Formation:	6.2W	49.0N	reversed samples	
	Caballeros Canyon	6.6W	21.5N	normal samples	

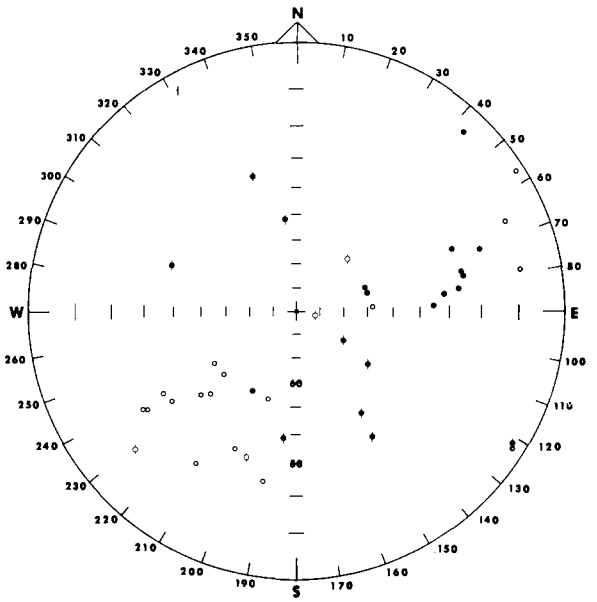


Fig. 5. Stereogram of the directions of natural remanent magnetization of the La Boca Formation, Los Caballeros Canyon. Points indicated by ϕ did not respond to thermal demagnetization and are not shown in Fig. 6.

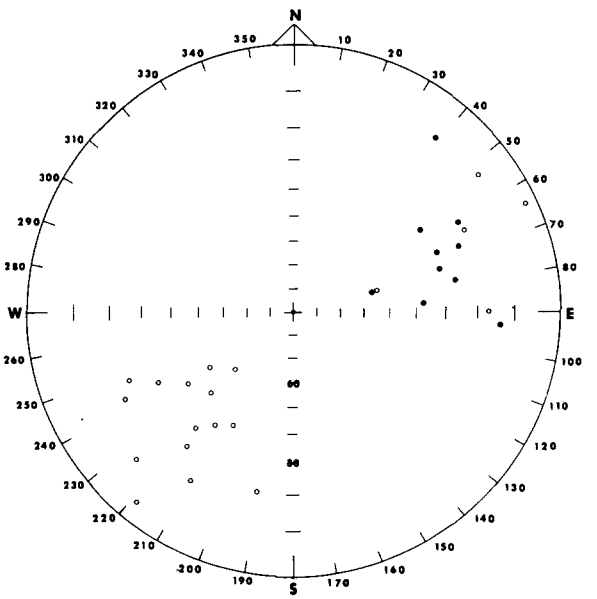


Fig. 6. Stereogram and the directions of magnetization of samples from the La Boca Formation, Caballeros Canyon after thermal cleaning at 450°C.

original Huizachal Formation near the village of that name. They crop out alongside the former main road and were described by Baker (1971). The section contains numerous conglomerate horizons which were not sampled and in the lower part of the section beds are affected by folding and faulting on a relatively small scale.

In the section it was not feasible to break down the samples into various beds, consequently all the measurements are combined to give a single mean. All the samples were thermally cleaned and give the maximum K value after treatment at 400°C. The individual measurements are figured in Fig. 8. The stability of remanence is illustrated by stable-end-point-direction method after As and Zijderveld (Fig. 7).

A second sequence of samples was collected in the Caballeros Canyon a little to the northwest of Cuidad Victoria where the beds demonstrably overlie beds of the La Boca Formation with strong angular unconformity. The samples were collected approximately 60 ft. above the base of the formation and represent a horizon within the La Joya Formation below the horizons collected near Huizachal, where the lower part of the section was predominantly conglomeratic. While the magnetization of these rocks appears to be stable, at least with respect to thermal demagnetization, the measurements appear at first sight to be widely scattered. If examined in stratigraphic order, however, they appear to be recording the anomalous behaviour of the magnetic field oscillating between

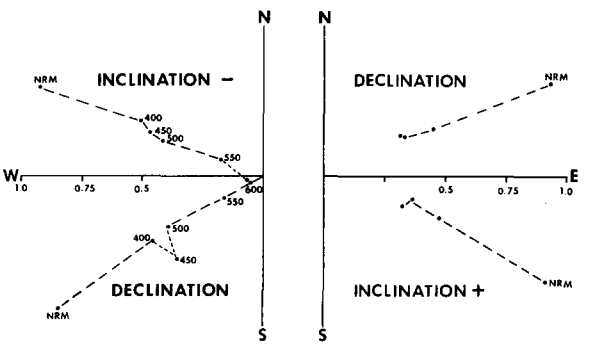


Fig. 7. Stable end-point curves showing the change in declination and inclination during progressive thermal demagnetization. The intensity is normalized. Curves are from samples of the La Joya Formation. The declination is plotted in the horizontal plane, inclination in the vertical, meridional section.

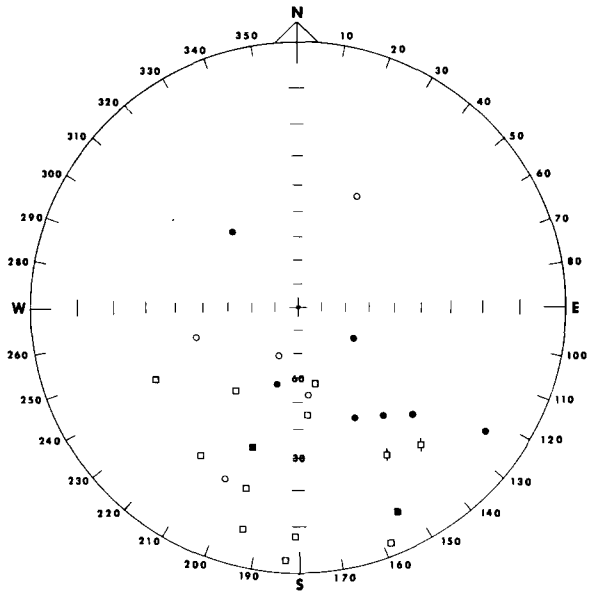


Fig. 8. Stereogram illustrating the directions of magnetization of individual cores from the La Joya Formation near Cuidad Victoria, Tamaulipas, NRM Results shown by ●○ and after thermal demagnetization at 450–500°C by ■□. Samples for which no NRM values are plotted are identified by a ϕ symbol.

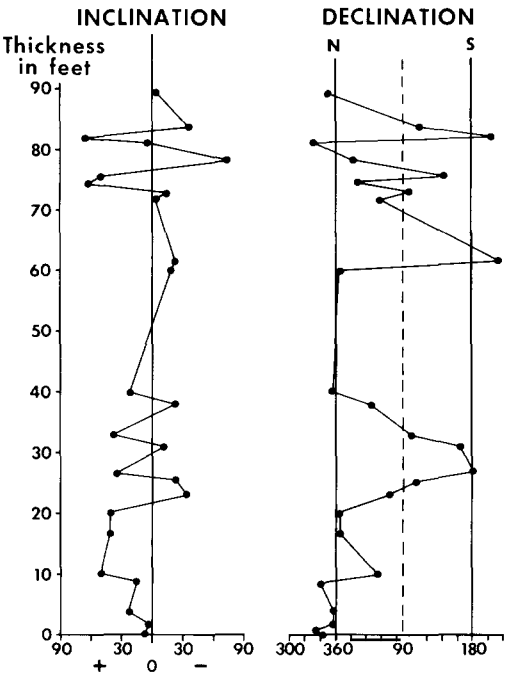


Fig. 9. A plot showing the changes in declination and inclination in samples collected from a 90-ft. thickness of La Joya Formation red beds collected in Los Caballeros Canyon, some 20 ft. above the base of the formation.

the reverse direction recorded from the rocks in the Huizachal section and a direction 180° away (Fig. 9). There appears to be two such oscillations recorded, the earlier about a N–S axis and a younger which has a more SE–NW trend (Fig. 10).

An alternative interpretation suggested by the work of Larson and Walker (1975) is that there may be more than one mineral carrier of remanence with the intermediate directions being the resultant of opposed directions associated with different mineral phases or equally hard primary and secondary components of magnetization. Although there was little evidence of multiple phases in the thermal demagnetization this possibility should not be excluded.

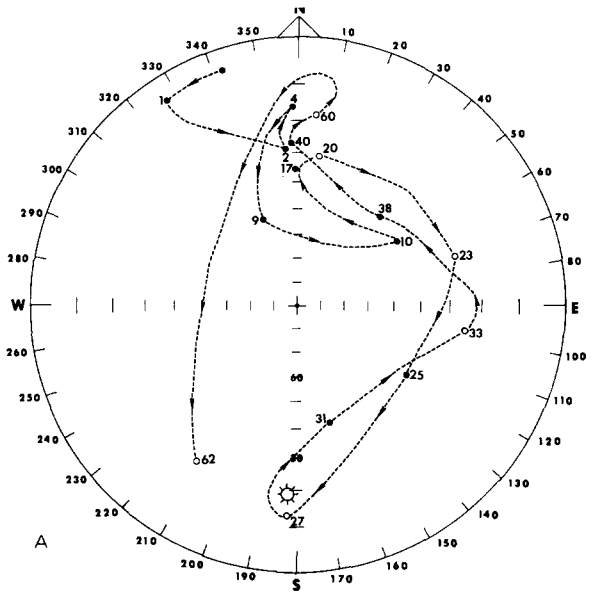
3.5. Difunta Group: Cerro Huerto and Las Imagenes Formation

The stratigraphy of the Difunta Group has recently been revised (McBride et al., 1974). Samples were obtained from the Cerro Huerta Formation and the Las Imagenes Formation.

They were sampled at three locations, on the slopes of Cerro Huerta southwest of Saltillo, in a section near La Escondida some 20 km west of Saltillo, and the thickest section of all, on the slopes of Sierra de Guijardo immediately north of the town.

On the slopes of Cerro Huerta, the beds of red sandstone alternate with non-red beds, and at La Escondida again sampling was restricted to discrete red sandstone beds separated by friable shale. The sections north of Saltillo contained a much higher proportion of red material. The principal section, immediately north of Saltillo, consisted mainly of red shale with intervening sandier horizons. The hillside was covered by a rubble made up of small fragments of shale weathered out along joints and fissures. Sampling was therefore restricted to the more massive, sandier horizons. These horizons were not always laterally continuous. Under these circumstances the number of cores collected from thicknesses usually up to 1 m was variable but seldom more than two or three. Losses due to cores crumbling or fragmenting prior to, or during treatment were much higher than normal.

Pilot samples were thermally demagnetized at 100° intervals up to 400°C and then at 50° intervals to



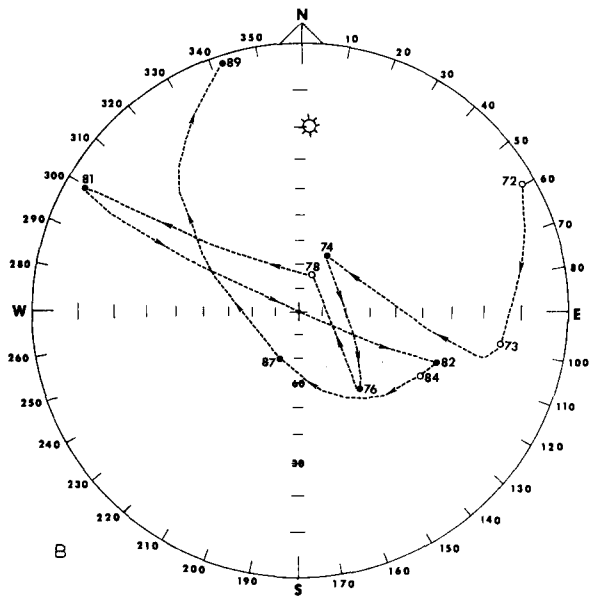


Fig. 10A and B. Stereographic plots of the cleaned directions of magnetization of samples of the La Joya Formation collected in Los Caballeros canyon. The individual points are connected in stratigraphic sequence from bottom to top, the numbers reflect the cumulative stratigraphic thickness above the initial point (itself some 20 ft. above the base of the formation). The points marked and represent the mean and the reverse directions of the La Joya samples collected near Huizachal.

550°C with the best results found after demagnetization at temperatures in the range 450–550°C. All the remaining samples were then demagnetized and measured over the 400–500°C range (Fig. 11). Means were calculated for each band or horizon selecting the temperature interval giving least scatter. These means were then used to compute a mean for the section.

At La Escondida where the Las Imagenes Formation is exposed the only reversed directions of magnetization were found in the samples collected. The scatter within each site was considerable, suggesting that at temperatures as high as 550°C not all the secondary magnetization has been removed. It is also possible that the results from the few horizons measured were insufficient to average out the secular variation. The data, therefore, are used only as evidence of the existence of at least one horizon of reversed magnetization within the Maestrichtian. Fortunately, the results from the normally magnetized beds of the Cerro Huerta Formation were better grouped and changes during demagnetization up to 550°C were smaller. In a recently published

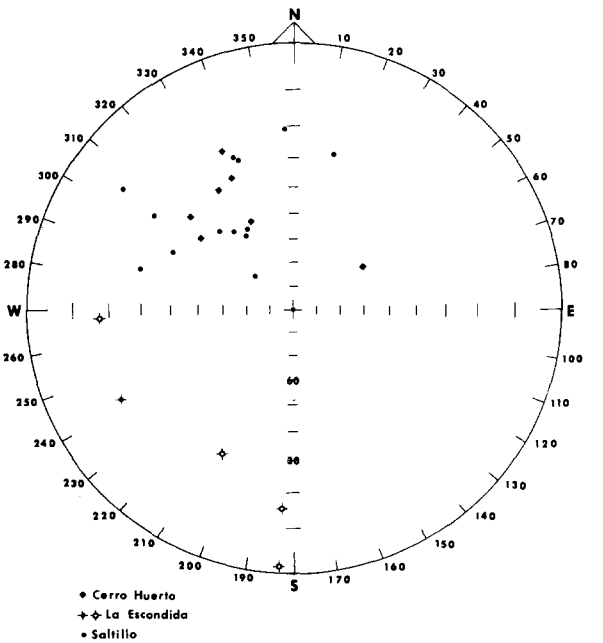


Fig. 11. Stereogram illustrating the site mean directions of magnetization after thermal demagnetization of rocks from the Cerro Huerto and Las Imagenes Formations, Difunta Group from the vicinity of Saltillo, Coahuila.

study, the polarity succession of the Maestrichtian, Keating et al. (1975) indicate several polarity changes. This suggests that the thick section on the Sierra de Guijardo represents a relatively short time, or that in sampling all reversely magnetized horizons were missed.

4. Discussion of results

From the mean data assembled it is possible to compute a Cretaceous (Maestrichtian) virtual pole based upon the results from the Cerro Huerto and Sierra de Guijardo sections. It is also possible to calculate an Early Jurassic virtual pole for the La Joya Formation and an ?Upper Triassic virtual pole calculated from the Nazas Formation. The virtual pole from the "Huizachal Formation" south of La Murella lies between the major grouping of Triassic and Cretaceous virtual poles taken from Hicken et al. (see Fig. 12), so that it is suggestive that the pole in question reflects beds younger than the Nazas Formation, and of a probable Jurassic age not inconsistent with the age assignment, although the pole is different from that calculated for the La Joya Formation.

The virtual pole calculated for the La Boca Formation provides something of a puzzle. It could have been hoped that a palaeomagnetic result would help resolve the stratigraphic problem of its age assignment, yet the pole calculated is not only inconsistent with the La Joya and Huizachal result, but also with every Mesozoic and Late Paleozoic result from North America.

Since the Cretaceous and ?Triassic results are not significantly different from other results from the North American craton, and the Huizachal pole is close to the position anticipated, given the age assigned to it, it is reasonable to conclude that during Mesozoic times the North American craton extended south of the Rio Grande to 23°N and by inference to at least as far south as the 19th parallel. This implies that either the La Boca is much older than has previously been assumed, and from the virtual pole position it is difficult to assign an age younger than Precambrian or Early Palaeozoic, or that the region during the course of folding underwent some local rotational movement. It is difficult to associate such a motion with the curvature of the Sierra Madre Oriental for the rotation which would have to move the southern end of the Sierra Madre Oriental some 40–50° in a clockwise direction, would also have to have been completed between Late Triassic and Early Jurassic times. Purely local movement, however, cannot be ruled out, although geologically it appears unlikely. As the beds are mapped as overlying a Paleozoic section (Fries et al., 1962) it is clear that a reinvestigation of the field relationships is needed to resolve the problem.

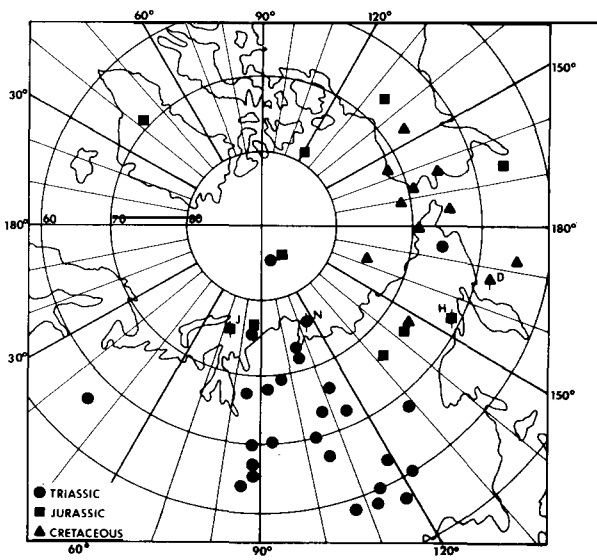


Fig. 12. Paleomagnetic pole positions for the Mesozoic of North America. Solid symbols represent the virtual pole positions from Mexico, open symbols; the class (e.g., the Franciscan results). For convenience the general region in which the Triassic and Cretaceous virtual poles lie have been outlined: Triassic, Jurassic and Cretaceous.

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